



THOMAS INTERNATIONAL MUSIC ACADEMY (TIMA)

MASTER TECHNICAL SPECIFICATIONS & SYSTEM ARCHITECTURE



"Sing and make music from your heart to the Lord." — Ephesians 5:19

1. Technical Architecture & Tech Stack

Layer	Technology	Key Specifications
Frontend Client	Vanilla JS (ES2024), Canvas, PWA	Single-page reactive UI, Web Audio API synth, touch crop engine, GSAP animations
Backend Server	Node.js (>= 22.5.0 Standard Library)	Zero external npm dependencies, native http, native crypto, native sqlite driver
Database	SQLite 3 (WAL Mode)	PRAGMA journal_mode = WAL, transactional ACID guarantees, auto-migration
Push Engine	Web Push (RFC 8291 + RFC 8292)	VAPID authentication, aes128gcm payload encryption, background service worker
Edge Backend	Cloudflare Worker / D1	Optional serverless push gateway using Web Crypto API and KV/D1 storage

2. Database Schema & Tables Overview

The system runs on a self-healing SQLite database (`sir\_slot.db`) containing 11 production tables:

- **students:** `id, name, mobile, pin\_hash, salt, batch\_id, time\_slot, slot\_type, group\_members, instruments, photo, ...`
- **coaches:** `id, name ('Thomas Sir'), pin\_hash, salt, mobile ('9848173025'), created\_at`
- **attendance:** `id, student\_id, date\_key, status, member\_statuses, note, marked\_at`
- **requests:** `id, name, mobile, photo, instrument, preferred\_days, preferred\_time, status, ...`
- **performance_logs, bookings, makeup_slots, push_subscriptions, sent_notifications, config, activity_logs**

3. Deployment & Zero-Configuration Execution

Run the server directly without installing any npm packages:

node server.js

The app binds to **0.0.0.0:3000** for seamless local Wi-Fi and mobile device access.